

BSc

BIOMEDICAL ENGINEERING

Biomedical Engineering at ACity

Aimed at revolutionizing medical care in Africa, students of ACity's top-end BSc Biomedical Engineering program will be dedicated to the design, analysis and modeling, programming, integration of AI, Biomedical data and informatics to develop medical systems, devices and equipment to manage, and improve healthcare delivery.

The creatively carved 70% practical program blends elements of Artificial Intelligence, Biomedical and Robotics to open new career avenues for students in medical devices development, medical data informatics, tissue engineering, rehabilitation medicine and the design of advanced prosthetics, especially for emerging markets.



Advancing Healthcare with Technology

Given the current global health crisis, it has become more incumbent to marry health science and engineering to expedite the solution-finding process of health-related issues.

The world is now experiencing the innovation of cutting-edge technologies to tackle the challenges faced in the health sector from the wheelchair that aids movement to the pain relievers that rests on our tables. The world is redefining quality human health with the birth of biomedical engineers who harness their creative skills and knowledge to reshape the areas of biology and medicine.

Top Careers In Biomedical Engineering

There are many career options to be considered in the field of biomedical engineering! You can save lives too by being any of these after 4 years as a Biomed Engineering student in ACity.

- ⑤ Biosignal and Medical Image Processing Specialist
- ⑤ Medical Technologist Biosafety Manager
- ⑤ Product Development Engineer
- ⑤ Tissue and Organ Engineer
- ⑤ Biomechanics Specialist
- ⑤ Biomedical Researcher

Why ACity?

Every ACitizen has the opportunity to do a **work placement** every vacation

Home to students from over
23 countries
across Africa

70%
practical lessons

Experienced Faculty
with Industry Expertise

Student-To-Faculty Ratio
5:1

A fully digitized campus with the fastest
internet speed
in a Ghanaian university

Houses a
Digital Makerspace
and a **Technology & Entrepreneurial Center**
for **Creative freedom & Innovation**

Our Unique Learning Pillars



Experiential Learning

Hands-on learning to prepare students to readily apply concepts, to easily integrate into the workspace.



Contextual Learning

Solving real grass-root problems to expose students to the local context and develop empathy towards the continent's progress.



Unified Learning

A project-based approach that combines concepts across courses to connect the dots and enable unified learning.



Extensional Learning

Arms students with a viable toolkit to help them confront real-life issues, they may not have encountered during their academic life, squarely.

Entry Requirements

WASSCE/NECO

Minimum C6 in 6 subjects including 3 core subjects (English, Mathematics, Integrated/General Science) and 3 elective subjects. (Physics, Elective Mathematics and Biology)

SSCE

Minimum D or a pass in 6 subjects including 3 core subjects (English, Mathematics, Integrated/General Science) and 3 elective subjects. (Physics, Elective Mathematics and Biology)

IGCSE O-LEVEL & A-LEVEL

Minimum of 5 credit passes in the IGCSE/O-Levels (English and Maths mandatory) and 3 passes in the A-Levels. (Elective/Add/Further Maths and Physics mandatory)

ENGLISH IB

Minimum of 5 credit passes in the IGCSE/O-Levels (English and Maths mandatory) and a minimum score of 4 points in 3 Higher Level (HL) subjects. (Elective/Add/Further Maths and Physics mandatory)

FRENCH IB

Minimum of 50% overall average pass. (subject to approval NAB) (Maths, English and Physics mandatory)

AMERICAN HIGH SCHOOL

Minimum GPA of 3.0 (Maths, English and Physics mandatory)

COURSE OUTLINE

SEMESTER 1

Course Name

Communication Skills

French Language

Fundamentals of Innovation and Entrepreneurship (FIE) Seminar I

Introduction to Programming with Python

Technology and Society

Physics I: Mechanics

Pre-Calculus (with MATLAB)

Introduction to Biomedical Engineering

SEMESTER 2

Course Name

Fundamentals of Innovation and Entrepreneurship (FIE) Seminar II

Logic and Critical Thinking

Text and Meaning

Introduction to Multidisciplinary Design

Programming in C

Physics II: Electromagnetic Field Theory

Principles of Chemistry I

Analytic Geometry and Calculus I (with MATLAB)

Introduction to Electrical and Electronics Engineering

SEMESTER 3

Course Name

Fundamentals of Innovation and Entrepreneurship (FIE) I

Leadership Seminar I

Physics III: Optics

Principles of Chemistry II

Analytic Geometry and Calculus II (with MATLAB)

Circuit Theory

Computer Aided Design and Manufacturing (CAD and CAM)

Engineering Thermodynamics

SEMESTER 4

Course Name

African Studies

Fundamentals of Innovation and Entrepreneurship (FIE) II

Applied Linear Algebra (with MATLAB)

Probability, Statistics and Reliability (with MATLAB)

Analogue and Digital Electronics

Human Anatomy and Physiology

Introduction to Artificial Intelligence and Robotics

Introduction to Organic Chemistry

SEMESTER 5

Course Name

Leadership Seminar II

System Dynamics

Differential Equations
(with MATLAB)

Numerical Methods
(with MATLAB)

Biochemistry

Cell and Molecular Biology

Sensor and Bioinstrumentation I

Signals and Systems

SEMESTER 6

Course Name

Project Management, Engineering
Economics and Risk Analysis

Biomaterials

Biomedical Imaging and Sensing

Digital Signal Processing

Industry Internship

Research Methodology

Sensor and Bioinstrumentation II

SEMESTER 7

Course Name

Biomechanics

Biomedical Data Analysis with AI

Medical Devices and Technologies

Project Phase I

Quantitative Physiology

Rehabilitation Engineering

SEMESTER 8

Course Name

Environmental Science and
Engineering

Professional Ethics and Values

Advances in Imaging Technologies:
Neuroimaging

Medical Image Processing

Project Phase II

Real-Time Digital Signal Processing